

LOAD op = 2'b00 Memory → PE register

19:18	17:8	7:5	4:2	1:0
rsvd	addr [9:0]	(x)	rd	00
2b	10b	3b	3b	2b

R-TYPE op = 2'b01 rd = rs1 <alu_op> rs2

19:18	17:15	14:12	11:8	7:5	4:2	1:0
rsvd	alu_op	rs2	(rsvd)	rs1	rd	01
2b	3b	3b	4b	3b	3b	2b

STORE op = 2'b10 PE register → Memory

19:18	17:8	7:5	4:2	1:0
rsvd	addr [9:0]	rs1	(x)	10
2b	10b	3b	3b	2b

NEWS op = 2'b11 rd = rs1 <alu_op> neighbour(dir), rs2 drives NEWS out

19:18	17:15	14:12	11:10	9:8	7:5	4:2	1:0
rsvd	alu_op	rs2	dir	(rsvd)	rs1	rd	11
2b	3b	3b	2b	2b	3b	3b	2b